

CSCE 20004 – Programming Foundations I
Programming Project 1
Due Date: Friday, 08/29/2025, 11:59pm

1. Problem Statement:

The goal of this programming assignment is to give students experience performing numerical calculations and using input/output in C++. You were hired as head chef at a restaurant. Your first task will be to scale American recipes for different serving sizes and convert ingredient costs to metric currency.

Your program should do the following:

- Prompt the user for how many servings your recipe makes. For example, the number of servings can be 1, 2.5, 10, etc.
- Prompt the user for how many servings you want to scale up or down to. For example, scaling up to 20 servings or down to 2 servings.
- Prompt the user for the cost of your ingredients in USD.

Calculate the cost per serving in euros. You will need to use the US Dollar to EUR exchange rate (1 USD is equivalent to 0.85 EUR).

2. Design:

Your first design task is to decide how you want to do the program input/output. What messages you want to print, how to read inputs, how to write outputs. There is no “right way” or “wrong way” to do this but you want your program to be easy to use, and the outputs to be easy to understand.

Your second design task is to decide the formulas needed to scale a recipe a new number of servings. For example, if you’re increasing from 5 to 10 servings, you’d multiply each ingredient by 2 and translate them into C++ code. You should test your formulas and calculations by hand to verify that your code is correct.

Your final design task is to decide what formulas are needed to convert the cost of ingredients from USD to EUR and cost per serving in USD. You should test your formulas and calculations by hand to verify that your code is correct.

3. Implementation:

Since you are starting with a “blank piece of paper” to implement this project, it is very important to develop your code incrementally writing comments, adding code, compiling, debugging, a little bit at a time. This way, you always have a program that “does something” even if it is not complete.

As a first step, it is always a good idea to start with an empty main function, and add the code to read input data from the user, and then simply print these values back out again. Once this part is working, you can start performing calculations with the input data.

4. Testing:

Test your program with a variety of inputs to check that it operates correctly for all of the requirements listed above.

Once your program is running correctly, you should run it several times with different values for dimensions, costs, and budget. Include the results of your experiments in your project report.

You are NOT required to add error checking in this program, but it is always good to test a program to see what happens to your program if the user inputs unexpected data (like “hello mom”). You should cut/paste these results into your project report to document what your program does in these cases.

5. Documentation:

When you have completed your C++ program, write a short report using the “Programming Project Report Template” describing what the objectives were, what you did, and the status

of the program. Does your program work properly for all test cases? Are there any known problems? Save this project report in a separate document to be submitted electronically.

6. Sample Outputs:

Sample 1 -

```
Congratulations on getting hired as chef! Let's work on scaling
your American recipes to metric measurements and ingredient cost
s for different serving sizes.
```

```
Welcome to the Recipe Scaling Program!
Please enter the following details:
```

```
How many servings does your recipe make? 5
How many servings do you want to scale it to? 10
What is the cost of your ingredients in USD? 2.50
Cost per serving in USD: $0.50
Cost of ingredients in EUR: €2.12
```

```
To scale your recipe from 5.00 servings to 10.00 servings, you w
ill need to multiply each ingredient by 2.00.
Now, let's get started with your recipe!
```

Sample 2 -

```
Congratulations on getting hired as chef! Let's work on scaling
your American recipes to metric measurements and ingredient cost
s for different serving sizes.
```

```
Welcome to the Recipe Scaling Program!
Please enter the following details:
```

```
How many servings does your recipe make? 10
How many servings do you want to scale it to? 4
What is the cost of your ingredients in USD? 10
Cost per serving in USD: $1.00
Cost of ingredients in EUR: €8.50
```

```
To scale your recipe from 10.00 servings to 4.00 servings, you w
ill need to multiply each ingredient by 0.40.
Now, let's get started with your recipe!
```

7. Project Submission:

In this class, we will be using electronic project submission to make sure that all students hand their programming projects and labs on time, and to perform automatic plagiarism analysis of all programs that are submitted.

When you have completed the tasks above, go to Blackboard to upload your documentation (a single docx or pdf file), and your C++ program (a single cpp file). Do NOT upload an executable version of your program.

The dates on your electronic submission will be used to verify that you met the due date above. All late projects will receive reduced credit:

10% off if less than 1 day late,
20% off if less than 2 days late, 30% off
if less than 3 days late, no credit if more
than 3 days late.

You will receive partial credit for all programs that compile even if they do not meet all program requirements, so handing projects in on time is highly recommended.

8. Academic Honesty Statement:

Students are expected to submit their own work on all programming projects, unless group projects have been explicitly assigned. Students are NOT allowed to distribute code to each other, or copy code from another individual or website. Students ARE allowed to use any materials on the class website, or in the textbook, or ask the instructor and/or GTAs for assistance.

This course will be using highly effective program comparison software to calculate the similarity of all programs to each other, and to homework assignments from previous semesters. Please do not be tempted to plagiarize from another student.

Violations of the policies above will be reported to the Provost's office and may result in a ZERO on the programming project, an F in the class, or suspension from the university, depending on the severity of the violation and any history of prior violations.